

Googlantir
A Zero-Shot Time-Series Forecasting Application
Based on Google Research TimesFM Architecture

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Abstract

This report presents Googlantir, a web-based application designed for financial time-series forecasting. The system leverages TimesFM, a foundation model developed by Google Research, to generate probabilistic predictions without requiring specific training or fine-tuning. The application provides an accessible interface for exploring zero-shot forecasting capabilities across multiple temporal granularities, offering configurable context windows and forecast horizons with specific uncertainty bounds.

1 Why TimesFM ?

Googlantir is a forecasting tool implemented as a Streamlit-based web application in Python. The system is designed to perform predictive analysis on financial time series data retrieved from public markets via Yahoo Finance API. The primary distinguishing characteristic of TimesFM (Time Series Foundation Model), is its zero-shot capability, which, eliminates the conventional requirement for model retraining or fine-tuning on target-specific datasets. This property enables immediate deployment, without incurring in a computational heavy load or in a data requirements issue, typically associated with supervised learning approaches. The model operates effectively in a local execution environment, requiring no external API calls for inference once the pre-trained weights are loaded.

2 System Architecture

The application comprises three primary components structured in a modular architecture. **User Interface Layer:** The front-end is implemented using Streamlit, providing ticker symbol

input with validation, three operational modes for different temporal resolutions (hourly, daily, weekly), interactive parameter controls for context length and forecast horizon, and real-time visualization of predictions with confidence intervals. The design incorporates informational sections explaining the tradeoffs in time-series forecasting, particularly the noise-bias balance in context window selection. **Data Acquisition Module:** The data layer utilizes the `yfinance` library to retrieve historical price data from Yahoo Finance, handling multi-granularity data fetching (1-hour, 1-day, 1-week intervals), timezone normalization for cross-market compatibility, differentiation between cryptocurrency (24/7) and equity (trading hours) data streams, and validation of data sufficiency relative to model requirements.

2.1 Forecasting Engine

The core prediction logic is encapsulated in a separate module that interfaces with the TimesFM model. Key operations include:

- Model instantiation with configurable context and horizon parameters
- Input normalization using standardization (zero mean, unit variance)
- Inference execution with quantile regression for uncertainty quantification
- Post-processing to generate point forecasts and confidence bands (10th, 30th, 70th, 90th percentiles)

3 Technical Implementation Details

3.1 TimesFM Model Characteristics

TimesFM is a decoder-only transformer architecture pre-trained on approximately 100 billion real-world and synthetic time-series data points. The model exhibits some fundamental properties, that enable an easy implementation:

- **Zero-shot generalization**: The model performs inference on unseen time series without gradient updates or retraining. This capability stems from its pre-training on a diverse corpus spanning multiple domains, frequencies, and statistical characteristics.
- **Local execution**: The model operates entirely within the local computational environment. Once the 200-million-parameter checkpoint is downloaded, no external network requests are required for inference.
- **Frequency-agnostic processing**: The architecture takes a frequency token as input, allowing it to handle time series with different sampling rates (hourly, daily, weekly) in a consistent way without changing the model structure.
- **Probabilistic outputs**: Beyond point predictions, the model generates a continuous quantile function, allowing for uncertainty estimation at arbitrary confidence levels.

3.2 Challenges encountered and Solutions

Temporal Discontinuities in Financial Data Financial market time is not truly "continuous": there are trading sessions, closures, and weekends. At hourly or daily resolution, generating timestamps naively would create artificial gaps. To avoid this, the implementation first distinguishes between cryptocurrencies (continuous trading) and equities (discrete trading sessions). For equity data, it then filters out non-trading hours (pre-market and after-hours) and excludes weekends. Finally, Plotly range breaks are used so that charts do not look stretched or visually distorted by these discontinuities.

Data Availability Constraints The Yahoo Finance API introduces practical constraints, including implicit rate limits and limits on the historical depth that can be fetched reliably.

The application mitigates these issues by mapping short user-selected periods to slightly larger data fetches (e.g., requesting "2 weeks" triggers a 1-month download, followed by truncation). Before running the model, it also checks that the series contains enough observations (at least 35). When something goes wrong, the system provides informative error messages that clearly separate invalid tickers from cases where the historical data is simply insufficient.

Model Recompilation Overhead In the TimesFM implementation, changing the context length or the forecast horizon can trigger JIT compilation, which can introduce noticeable latency. To keep the interface responsive, model instances are cached via the `@st.cache_resource` decorator, using cache keys that include both context length and forecast horizon. The initial compilation delay is also made explicit to the user through loading indicators.

4 Methodological Considerations

4.1 Context-Horizon Tradeoff

The application enforces no hard constraints on parameter selection but provides guidance through educational components. The fundamental tradeoff is:

- **Short context**: High sensitivity to recent dynamics, increased variance, reduced ability to capture long-term periodicity
- **Long context**: Better seasonality detection, reduced noise sensitivity, potential staleness if underlying regime has shifted

It is recommended by Google to maintain a 3:1 to 4:1 context-to-horizon ratio emerges from empirical observations in time-series forecasting literature, balancing information content against overfitting risks.

5 Limitations and Future Work

The current implementation exhibits several constraints:

- **Univariate focus**: The model processes only closing prices, ignoring volume, volatility, and other market signals
- **No exogenous variables**: Macroeconomic indicators, news sentiment, and earnings data are not incorporated

6 Conclusion

Googlantir is a strong example of "lightweight AI": a foundation model that, despite its substantial size, can still be used and deployed with relative ease. Zero-shot architectures are an important step toward making these technologies more streamlined, and TimesFM is a clear demonstration of this trend. At the same time, the model features a modular design, which avoids the need for structural changes when working with time series sampled at very different rates. This property, combined with local execution, makes the system a practical tool for rapid prototyping and for educational exploration of time-series forecasting principles. The hope is that more and more models of this kind—compact, accessible, and efficient—will become widely available.

Repository: `https://github.com/npasi/Googlantir`
Technologies: Python, Streamlit, PyTorch, TimesFM, yfinance, Plotly